

b)
$$\begin{bmatrix} \square & * & * & * & * \\ 0 & 0 & \square & * & * \\ 0 & 0 & 0 & \square & * \end{bmatrix}$$

The corresponding equations are:

1. $a_1x_1 + b_1x_2 + c_1x_3 + d_1x_4 = e_1$
2. $c_2x_3 + d_2x_4 = e_2$
3. $d_3x_4 = e_3$

$a_1 \neq 0$ $c_2 \neq 0$ $d_3 \neq 0$. From eqn 3, we get a unique value of x_4 .
 Substituting value of x_4 in eqn 2, we get a unique value of x_3 .
 Substituting value of x_3 and x_4 in eqn 1, we get a generalized value of x_1 in terms of x_2 . Column 2 has no pivots. $\therefore x_2$ is a free variable.

By Theorem 2, the system is consistent and has infinitely many solutions.

In Exercises 17 and 18, determine the values of h such that the matrix is the augmented matrix of a consistent linear system.

17.
$$\begin{bmatrix} 2 & 3 & h \\ 4 & 6 & 7 \end{bmatrix}$$

Step 1:

Row 2 = Row 2 - 2x Row 1

$$\begin{bmatrix} 2 & 3 & h \\ 0 & 0 & 7-2h \end{bmatrix}$$

Corresponding eqns:

- 1) $2x + 3y = h$
- 2) $0 = 7 - 2h$

By Theorem 2, the given system will be consistent if and only if $7 - 2h = 0 \Rightarrow h = 3.5$

18.
$$\begin{bmatrix} 1 & -3 & -2 \\ 5 & h & -7 \end{bmatrix}$$

Step 1:

Row 2 = Row 2 - 5x Row 1

$$\begin{bmatrix} 1 & -3 & -2 \\ 0 & h+15 & 3 \end{bmatrix}$$

The corresponding eqns:

1. $x - 3y = -2$
2. $(h+15)y = 3$

By Theorem 2, the given system will be consistent if and only if LHS of eqn 2 $\neq 0$, i.e. $h \neq -15$.

In Exercises 19 and 20, choose h and K such that the system has a) no solution b) a unique solution, and c) many solutions. Give separate answers for each part.